

A Validated Reduced-Order Model for a Submerged Tube-Bundle Condenser Pool-Boiling Chamber, with Manifold-Resolved Condenser CFD and its Link to Boiling-Curve Performance

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Abstract

We report a validated gray-box reduced-order model for a closed subcooled pool-boiling chamber cooled by a partially submerged tube-bundle condenser, a platform aimed at two-phase immersion cooling of high-power electronics. Against **261 measured points** spanning two condenser geometries (33- and 42-tube), two chip surfaces (plain and microchannel), and four coolant inlet temperatures (20 °C to 50 °C), the model predicts chip temperature with an overall $R^2 = 0.938$, **RMSE** 4.34 K, and bias 0.89 K. The 33-tube plain configuration is reproduced best ($R^2 = 0.964$, 91 % of points within 5 K), while the 42-tube configurations are weaker and biased high by ~ 2 K. We then present three-dimensional, STEP-faithful single-phase OpenFOAM simulations of the coolant manifolds for all three condenser generations (33-, 42-, and 66-tube), which resolve the per-tube flow distribution and the pressure budget. The simulations show that distribution and pressure drop are governed by the *manifold feed geometry and the tube bore size*, not the tube count: a single centred port produces severe maldistribution while top-fed vertical manifolds distribute evenly, and wide bores render the bundle hydraulically near-invisible while narrow bores carry a measurable share of the drop. Finally we connect the two analyses: the coolant-side thermal resistance that the manifold CFD informs is exactly the quantity that sets the chamber saturation temperature and therefore the vertical position of the boiling curve. This framework rationalises why the model is most robust for the wide-bore, high-flow 33-tube condenser (negligible coolant-side resistance) and weakest for the narrow-bore 42-tube condenser (significant, distribution-sensitive coolant-side resistance).

1 Introduction

Direct two-phase immersion cooling is an attractive route to rejecting the rising heat fluxes of data-center processors. The platform studied here is a closed chamber in which a heated chip boils a subcooled dielectric pool; the vapour rises and condenses on a tube bundle that is partially submerged in the same pool and carries a single-phase coolant. The chamber is self-contained: the condenser sets the operating pressure (and hence the saturation temperature) by balancing the vapour generated at the chip against the vapour condensed at the bundle. Because the chip temperature at a given heat flux is the sum of the wall superheat and the saturation temperature, condenser performance feeds directly into the achievable junction temperature.

Table 1: Chip-temperature fit by configuration (in-sample). RMSE, MAE, bias and maximum absolute residual in kelvin.

Configuration	N	RMSE	MAE	Bias	R^2	$\max r $	within 5 K
33-tube plain	87	3.14	2.59	−0.99	0.964	8.4	91 %
33-tube micro	36	4.82	3.85	+0.88	0.867	10.2	67 %
42-tube plain	64	4.42	3.36	+2.00	0.900	8.1	66 %
42-tube micro	74	5.17	4.47	+2.16	0.927	10.6	65 %
Overall	261	4.34	3.48	+0.89	0.938	10.6	74%

This paper has three goals. First, we summarise a reduced-order model of the chamber and validate it against an extensive measurement set. Second, we use three-dimensional CFD to resolve the coolant-side flow distribution in the condenser manifolds, which has not previously been characterised for this apparatus. Third, and most importantly, we draw the physical link between the manifold-resolved condenser behaviour and the measured boiling-curve performance, giving a mechanistic explanation for where the reduced-order model is strong and where it is weak.

2 Apparatus and reduced-order model

The model is a gray-box: established two-phase and single-phase correlations carry the physics, and a small number of geometry-independent constants are fit once and then transferred across configurations. Nucleate boiling on the chip is represented in the Rohsenow framework with a surface–fluid constant C_{sf} ; the microchannel surface is treated with area-enhancement and the Cooke–Kandlikar description. The condenser side couples a tube-bundle condensation resistance to a single-phase internal coolant resistance (Gnielinski/laminar developing flow as appropriate) and an external bundle resistance. The chamber energy balance closes the loop: for a heat load Q , the saturation temperature follows

$$T_{\text{sat}} = T_{\text{cool}} + Q R_{\text{cond}}, \quad (1)$$

where R_{cond} is the total condenser thermal resistance and T_{cool} the coolant inlet temperature. The chip temperature is then

$$T_{\text{surf}} = T_{\text{sat}} + \Delta T_{\text{sat}}(q), \quad (2)$$

with the wall superheat ΔT_{sat} set by the boiling correlation. A key design property of this decomposition is that C_{sf} is a property of the chip surface and the fluid, not of the chamber: values fit on one condenser transfer to the other, which is what makes the model predictive rather than a per-configuration curve fit.

3 Experimental validation: model versus measurement

Figure 1 compares predicted and measured chip temperature for all 261 points. The predictions track the 1:1 line across the full 30 °C to 120 °C range, with most points inside a ± 5 K band. Table 1 reports the fit by configuration.

The boiling-curve view (Figure 2) is more physically revealing than the parity plot. For each configuration the measured heat flux is plotted against wall superheat $T_{\text{surf}} - T_{\text{sat}}$ (filled markers), with the model prediction overlaid (open markers), separated by coolant temperature. The model

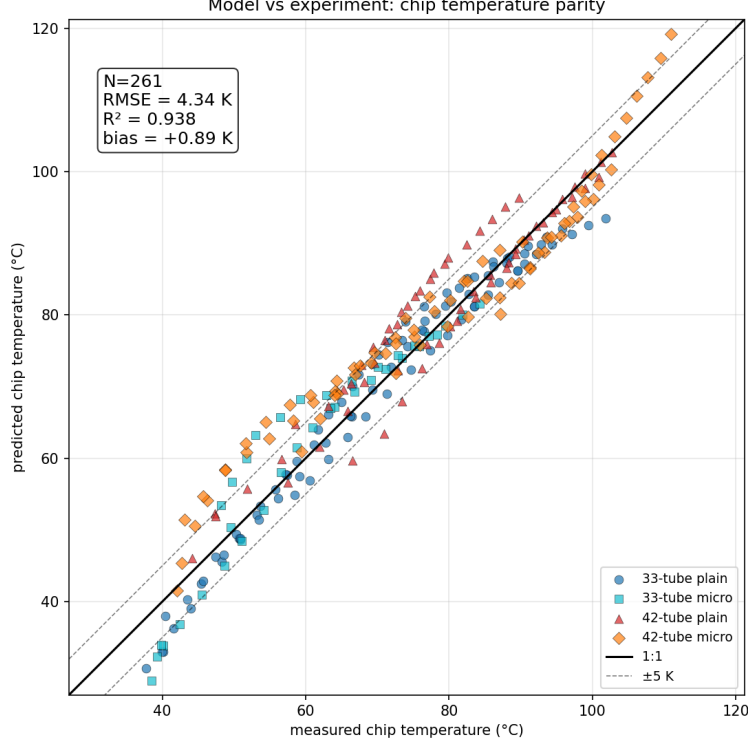


Figure 1: Predicted versus measured chip temperature for all 261 points, by configuration. The model achieves $R^2 = 0.938$ and RMSE 4.34 K. The 33-tube data (blue/cyan) hug the parity line; the 42-tube data (red/orange) sit slightly above it, reflecting a 2 K bias.

captures the entire coolant-temperature *family* of curves, including the rightward shift of the curve as the coolant warms. Two features stand out and recur in the residuals (Figure 3): the 33-tube panels track tightly, whereas the 42-tube panels show systematic over-prediction of the superheat (the open markers sit to the right of the filled markers).

Two diagnostics from the validation set are important for what follows. (i) The residuals show *no trend with heat flux*, which indicates the mechanistic correlations hold across the boiling regime rather than being tuned at a single operating point. (ii) The bias is *coolant-temperature dependent*: warm coolant (40 °C to 50 °C) is reproduced best (RMSE ~ 3.5 K) and the bias flips from 1.8 K at 20 °C to -1.4 K at 50 °C. A sign change in the bias with coolant temperature points to a residual in the *coolant-side* description, a clue that we return to in Section 5.

4 Condenser manifold CFD

The reduced-order condenser model lumps the coolant side into an effective resistance that assumes the coolant is distributed across the tubes. Whether that assumption holds, and what sets the pumping power, depends on the manifold hydraulics, which we resolve directly. For each condenser generation we built a STEP-faithful single-phase fluid domain (manifold cavities plus tube bores) from the production CAD and solved steady incompressible flow with **simpleFoam** (laminar, water properties at ~ 40 °C, 4 L min^{-1}). The three geometries differ in tube count, bore size, and crucially in how the coolant is fed:

- **33-tube** (bore ID 3.137 mm): a single coolant port on the bundle axis, feeding the manifold

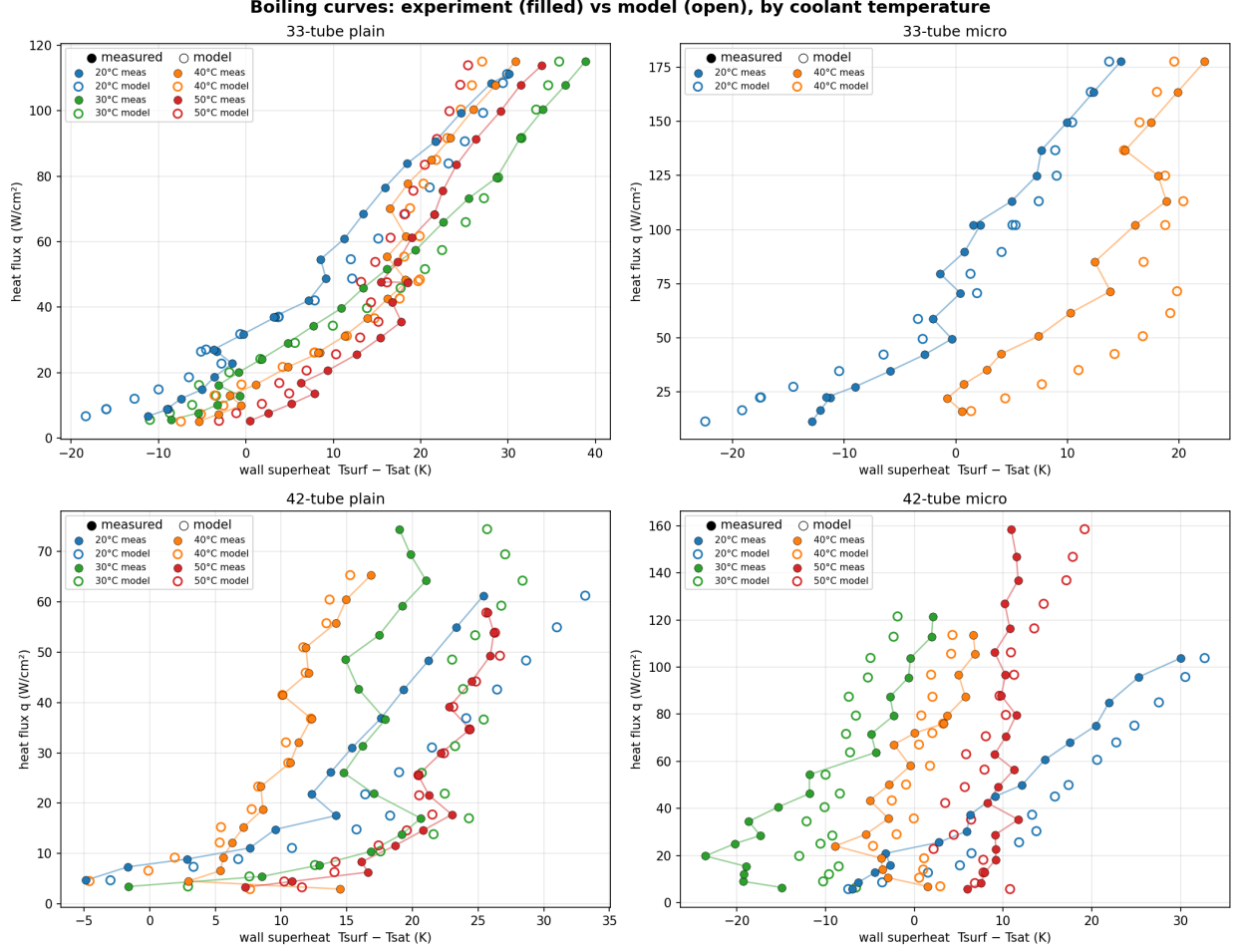


Figure 2: Boiling curves, experiment (filled) versus model (open), by coolant temperature. Heat flux versus wall superheat for each configuration. The model reproduces the coolant-temperature family; agreement is tight for the 33-tube and looser, with a rightward (over-prediction) offset, for the 42-tube.

centrally.

- **42-tube** (bore ID 1.39 mm): top-fed vertical manifolds at each tube end, coolant entering and leaving through ports on the manifold tops.
- **66-tube** (bore ID 1.6 mm, forecast geometry): single-pass, fed through a centreline port (modelled here with a simplified diffuser plenum).

4.1 Per-tube distribution

The distributions are strikingly different (Figure 4). The 33-tube centred port produces a *bullseye*: a single on-axis tube is over-fed to roughly eight times the mean while the rest are starved, giving a coefficient of variation $\text{CoV} = 131\%$. The 42-tube top-fed manifolds, in contrast, distribute the coolant evenly, $\text{CoV} = 10\%$, with only a mild bias toward the top-row tubes nearest the inlet. The simplified 66-tube centreline feed shows an intermediate, jet-dominated pattern ($\text{CoV} = 57\%$) in which a few central tubes intercept the inlet jet; we emphasise that this last figure is inflated by

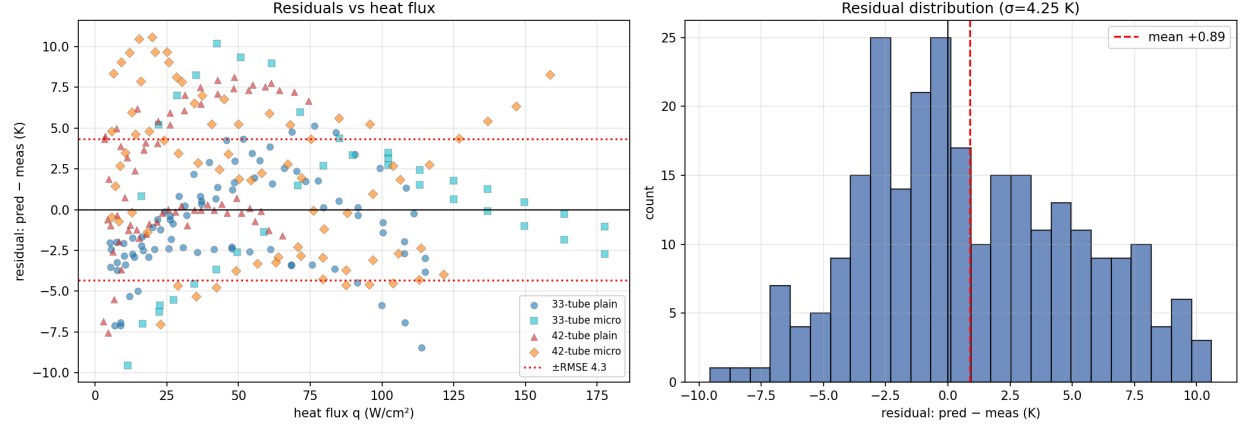


Figure 3: Left: residuals versus heat flux, showing no systematic trend with flux, the model is equally accurate at 10 and 170 W/cm². Right: residual distribution ($\sigma = 4.25$ K), centred near zero with a slight positive mean driven by the 42-tube.

Axial velocity at mid-tube plane (central spike, starved annulus)

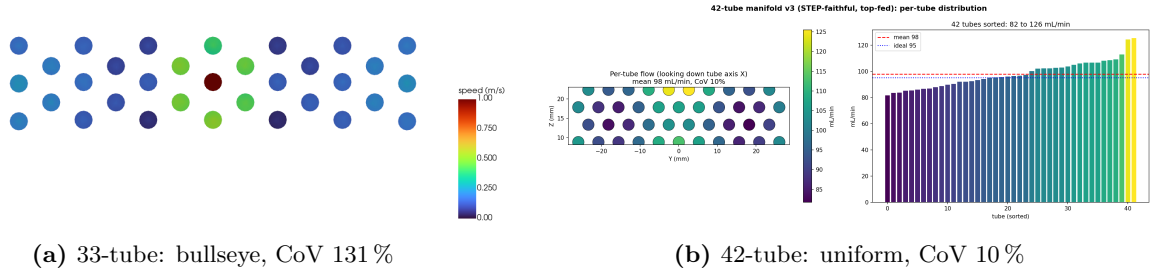


Figure 4: Per-tube coolant distribution from CFD. A single centred port (33-tube, left) starves all but the on-axis tube; top-fed vertical manifolds (42-tube, right) distribute evenly. Distribution is governed by the manifold feed geometry, not the tube count.

the simplified plenum and that the production manifold is a lofted diffuser designed precisely to spread that jet (Section 6). All cases conserve mass to within the discretisation error.

4.2 Pressure budget

The pressure analysis (Figure 5) reveals a second bore-size effect. In the wide-bore 33-tube condenser the pressure is essentially uniform through the entire bundle and the full drop occurs across the small outlet port contraction: the tube bundle is *hydraulically near-invisible* (the tubes contribute ~ 25 Pa of a ~ 10 kPa budget). In the narrow-bore 42-tube condenser the tubes carry a *measurable* share, ~ 1.4 kPa of distributed drop along the 62.6 mm length, although the port contractions still dominate. The 66-tube, with wide bores split among many parallel paths, again behaves like the 33-tube with the drop concentrated at the ports.

5 Linking condenser distribution to boiling-curve performance

The two analyses meet at the condenser thermal resistance R_{cond} in Eq. (1). The manifold CFD characterises the coolant side of that resistance; the boiling-curve validation measures its conse-

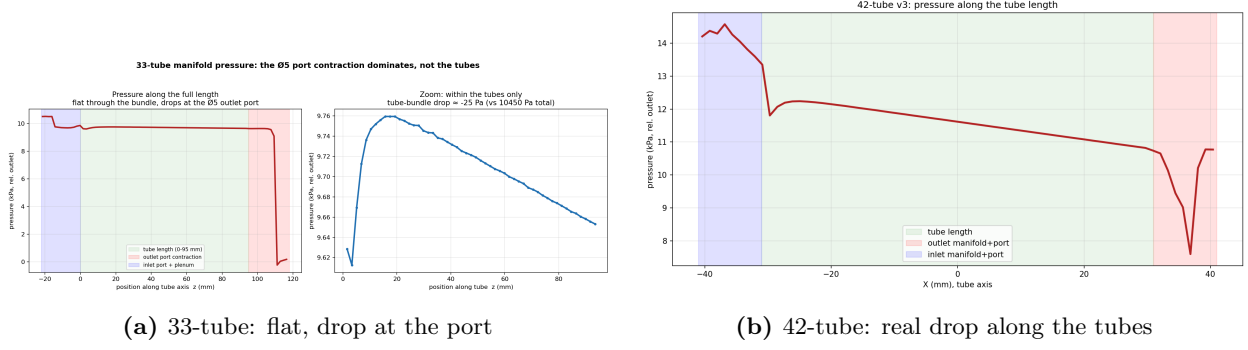


Figure 5: Pressure along the tube length. Wide bores (33-tube) leave the bundle hydraulically near-invisible with the drop concentrated at the port contraction; narrow bores (42-tube) carry a measurable distributed drop.

quence. The chain of causation is:

$$\underbrace{\text{manifold feed geometry}}_{\text{CFD}} \rightarrow \text{per-tube coolant flow} \rightarrow \underbrace{R_{\text{cond}}}_{\text{Eq. (1)}} \rightarrow T_{\text{sat}} \rightarrow \underbrace{T_{\text{surf}} = T_{\text{sat}} + \Delta T_{\text{sat}}(q)}_{\text{boiling curve}}. \quad (3)$$

A maldistributed manifold starves part of the bundle; starved tubes see their coolant warm toward the local saturation temperature, lose driving ΔT , and stop contributing effective area. The effective conductance UA falls, R_{cond} rises, and by Eq. (1) the chamber runs at a higher T_{sat} . Through Eq. (2) that lifts the entire boiling curve vertically: the chip runs hotter at every heat flux. Manifold distribution is therefore not a hydraulic curiosity, it is a direct lever on junction temperature.

The decisive subtlety is that the *magnitude* of this lever depends on how large the coolant-side resistance is relative to the rest of R_{cond} , and this is exactly what the bore-size pressure result foreshadows. The same property that makes the 33-tube bundle hydraulically invisible, wide bores and a high coolant flow (0.06 kg s^{-1}), also makes its coolant-side *thermal* resistance small: the coolant barely warms ($T_{\text{out}} - T_{\text{in}}$ is a fraction of a kelvin in the data), so even the severe bullseye maldistribution moves T_{sat} very little. The condenser is over-capacity, R_{cond} is dominated by terms insensitive to distribution, and the reduced-order model is correspondingly robust, hence the best fit of any configuration ($R^2 = 0.964$).

The 42-tube condenser sits at the opposite extreme. Its narrow bores (ID 1.39 mm) and lower coolant flow make the internal coolant resistance a significant fraction of R_{cond} , the same physics that makes its tubes carry a real pressure drop. The chamber saturation temperature is now genuinely sensitive to the per-tube flow and to the developing-flow internal heat-transfer description, and small errors in the coolant-side model translate directly into chip-temperature error. This offers a coherent, mechanistic explanation for the otherwise puzzling pairing in Table 1: the configuration with the *better* measured distribution (42-tube, CoV 10 %) is the *harder* one to model, because its performance lives where the coolant-side resistance matters. The coolant-temperature sign-flip in the 42-tube bias (Section 3) is consistent with this reading: a residual in the internal-resistance or subcooling treatment, amplified where the coolant side is not negligible.

This also clarifies how the manifold CFD should feed the reduced-order model. Where the coolant-side resistance is negligible (wide-bore, high-flow), a lumped, uniformly distributed UA is adequate and the CFD simply confirms it. Where it is significant (narrow-bore), the per-tube distribution from CFD can be folded into a distribution-weighted UA rather than a single bulk value, the most direct route to removing the 42-tube bias. For the forecast 66-tube, the lofted-

diffuser manifold is expected to restore near-uniform distribution; the simplified centreline-jet result should not be read as the apparatus prediction but as a demonstration of why the diffuser is needed.

6 Discussion and limitations

The validation is in-sample on the two built condensers; leave-one-coolant-out cross-validation (reported separately) shows stronger generalisation for the 33-tube than the 42-tube, consistent with the present fit. The manifold CFD is single-phase and steady, solved on tetrahedral meshes to a mesh-limited residual plateau, and is intended to characterise distribution and the pressure budget, not to be a converged high-fidelity solution. Two geometric simplifications are explicit: the 42-tube manifold was modelled with its as-built top ports, and the 66-tube inlet/outlet were simplified to a centreline-fed plenum rather than the production lofted diffuser. The 66-tube geometry is a forecast with no validation data, so all 66-tube results are exploratory. None of these simplifications affects the central qualitative findings, that distribution is set by feed geometry, that bore size sets whether the bundle carries hydraulic and thermal load, and that the coolant-side resistance is the hinge linking condenser CFD to boiling-curve performance.

7 Conclusions

1. A gray-box reduced-order model reproduces chip temperature across two condensers, two chip surfaces, and four coolant temperatures with $R^2 = 0.938$ and RMSE 4.34 K; the surface–fluid constant transfers across chambers, supporting a predictive rather than fitted construction.
2. STEP-faithful manifold CFD shows that per-tube distribution is governed by the feed geometry (a centred port gives CoV 131 %; top-fed manifolds give CoV 10 %) and that bore size sets the pressure budget (wide bores leave the bundle hydraulically near-invisible; narrow bores carry a real share).
3. The coolant-side thermal resistance links the two: it sets T_{sat} and hence the vertical position of the boiling curve. The reduced-order model is most robust where this resistance is negligible (wide-bore 33-tube) and weakest where it is significant (narrow-bore 42-tube), giving a mechanistic explanation for the configuration-dependent fit and a clear route to improvement: a distribution-weighted condenser conductance informed by the manifold CFD.

Data and reproducibility. All model code is consolidated in a single command-line tool; the CFD cases (geometry generators, meshes, run scripts, and solved fields) and the figure-generating scripts are archived with the project.